

claude-windows / ac48091f-a4f4-4268-b968-75205d9a50ff

Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\ac48091f-a4f4-4268-b968-75205d9a50ff\subagents\workflows\wf_6abc9156-811\agent-a06031ba3c4cc8757.jsonl`  
- SHA-256: `52e3b0bbe1be6a398007bfd490582c37024b05c65dc4568e09eaf3c4b18b264e`  
- Source modified: `2026-06-22T01:57:50+00:00`  
- Imported at: `2026-07-05T16:48:24+00:00`  
- project: `wf_6abc9156-811`  
- session_id: `ac48091f-a4f4-4268-b968-75205d9a50ff`
```

Transcript

1. user (2026-06-22T01:55:49.370Z)

You are scanning planning docs to surface ACCURATE, prioritized next steps for a co-founder.

REPO RULES:

```
- Engine repo at /c/Users/avidu/Projects/badminton-highlight-indexer: its working tree is on a  
STALE serving branch. Read DOCS + verify SHIPPED state from ENGINE origin/master ONLY:  
  git -C /c/Users/avidu/Projects/badminton-highlight-indexer show origin/master:docs/FILE.md  
(read a doc on master; if it errors, the doc may be branch-local ? fall back to the working-tree  
file and SAY SO)  
  git -C /c/Users/avidu/Projects/badminton-highlight-indexer log origin/master --oneline -60  
(what actually shipped ? use this to catch stale "TODO/next" claims)  
  git -C /c/Users/avidu/Projects/badminton-highlight-indexer grep -n "PATTERN" origin/master  
-- 'PATHGLOB'  
- khelsutra repo at /c/Users/avidu/Projects/khelsutra: on clean master ? use Read/Grep/Glob  
directly + git -C /c/Users/avidu/Projects/khelsutra log --oneline -40.
```

ANTI-SPIRAL (critical): before listing any "[gap]/net-new/TODO/next" item, VERIFY it hasn't already shipped ? cross-check the doc's claim against git log origin/master. A doc saying "X is next" is a HYPOTHESIS until you confirm X isn't already merged. Flag any stale doc claims you correct.

Context (already TRUE as of this session, do not re-list as "next"): the khelsutra ALPHA is DEPLOYED + live on a CPU droplet (engine + cloudflared tunnel + Cloudflare Access + SPA on Pages, real Gemini key working, full chain verified). DEPLOY_ALPHA fixes + the droplet kit are merged (khelsutra #78/#79/#81/#85). T2 (alpha deploy) is effectively DONE.

Return prioritized next items with honest status + git grounding.

```
TRACK: **rally-detection quality**. Read ENGINE origin/master `docs/NEXT_STEPS.md` (the quality-  
track pickup), `docs/HOW_RALLY_DETECTION_WORKS.md`, `docs/MULTI_SIGNAL_FUSION_PLAN.md`,  
`docs/EXPERIMENT_HARNESS.md`. Verify against git log: did the "held-shuttle Gate-A wiring" /  
multi-signal fusion / R2-metric / R4-end-rule items ship or are they still next? (The memory  
notes a Gate-A flip #146 that regressed ? #151; verify current state.)  
ANSWER: the cheapest quality win that's still open + the prioritized quality items. Ground in  
git.
```

2. assistant (2026-06-22T01:55:54.770Z)

I'll scan the rally-detection quality track docs on engine origin/master and verify shipped state against git log. Let me start by pulling the latest and reading the key docs.

3. user (2026-06-22T01:55:58.229Z)

---FETCH DONE---

a42c132 docs(pkg): track the batteries-included app plan (PACKAGING_PLAN.md) + index pointers (#309)

de2bf14 feat(pkg): P2e ? lazy-torch so the LIGHT tier imports with NO torch installed (default-OFF) (#308)

679e642 feat(pkg): P2d ? DB upgrade-safety (downgrade guard + backup-before-migrate + export) (default-OFF) (#307)

f01dc85 feat(db): v7 ? record the UI locale per job (PMF analytics) + count_jobs_by_locale (#306)

fa7ee3f feat(pkg): P2a ? single ffmpeg/ffprobe resolver (env-overridable; one fail-loud place) (default-OFF) (#305)

e99d0c3 feat(stitch): localize the reel watermark to the UI locale (+ OFL Noto fonts) (#304)

ca34787 feat(pkg): P2b ? friendly `indexer --app` launcher (boot localhost server + open the bundled UI) (#303)

4a06b8b feat(pkg): P1b ? bundle the KhelSutra SPA single-origin into the engine wheel (default-OFF) (#302)

be0c434 feat(pkg): P1a ? net-new console/wizard endpoints: /api/version, /api/capabilities, /api/videos, /api/reels (#300)

cb7394b feat(serving): T11 ? retention sweep for stale uploads + caches (#299)

c039b90 feat(pkg): P0e ? single-instance lock so a 2nd process can't corrupt in-flight job state (default-OFF) (#298)

e8bad40 feat(pkg): P0d ? declare scipy (undeclared serving dep) + license-scan & built-wheel CI lanes (#297)

3895963 feat(pkg): P0c ? config/profile-driven bind address + DNS-rebinding Host-header guard (default-OFF) (#296)

fc78fce feat(pkg): P0b ? redact secret-shaped fields from the unauthenticated GET /api/config (#295)

0c0e064 feat(serving): exposure-safety boot posture ? fail fast on a half-configured exposed engine (default-OFF) (#294)

82a28f4 feat(pkg): P0a ? resolved app-data HOME (RALLY_APP_HOME) for config/.env/DB/output (default-OFF) (#293)

2afbada fix(dryrun): install the watermark font in bootstrap + track the UI-trigger asks (#292)

7835f34 feat(gcp): always-install Ops Agent on GPU VMs + a cloud-GPU dry-run guide (#291)

aac7633 feat(serving): M9-auth ? Cloudflare Access edge-auth + per-tenant owner_id (default-OFF) (#290)

e68b583 fix(api): resolve a bucket-ingested video's source to a local file before stitching (#289)

43eccc1 feat(serving): M8 ? per-job cost ledger + versioned pricebook + /cost API (default-OFF) (#288)

8cdf65c feat(serving): M6 ? Cloud Run GPU dispatch shim + worker image (default-OFF, mocked) (#287)

b5af66e feat(serving): M5 ? truly-local profile per-profile startup checks + L4 profile-parity test (default-OFF) (#286)

fe4a112 docs(serving): resolve S8 design picks (D16/D17) + reconcile per-video P3 vs M3 (#285)

06ff4b7 test(api): lock gs://|gcs:// /api/process -> queryable API-DB video_id (B-P1 guard) (#284)

cee4baa feat(serving): M4 ? DeviceContext + native WASB CPU-fallback (device="auto", cuda stays strict) (#283)

2afd54f feat(serving): M3 ? configurable output/scratch base (de-hardcode output/, default-OFF) (#282)

41d45fd fix(server): drain async jobs under `python -m backend.main` (double-module hang) (#281)

e111add feat(serving): M2 ? URI-aware input (input_backend/input_root, local=identity) (#280)

ce22791 feat(serving): M1 ? deployment.profile + compute_target config (pure, additive, default-OFF) (#279)

f518e97 docs(serving): M0 APPROVED + lock D13-D15 (switchable profiles, Drive Northstar, consent-not-surprise) (#278)

49c082a docs(serving): lock the 5 M0 gating decisions (D7-D12) + add the data-flywheel harness (M11) (#277)

f592281 docs: point I18N_PLAN + UI_PROPOSAL (+ DOC_STATUS) at the khelsutra UX/l10n project (#275)

f6f15ca docs: isolate field-ops/PMF + Roora into archives/; carve a no-ML docs/data_pipeline/ (#276)

03fa951 docs: resolve DOC_STATUS findings #1 (per-video supersede) + #2 (NEXT_STEPS refresh) (#274)

aa5e9db docs: add DOC_STATUS register + archival due-diligence convention (#273)

bd42250 docs: spin up COMPUTE_DECOUPLED_SERVING subproject + ADR-014 (#272)

4957b72 fix(worker): configure logging + explain a 0-segment infer (the cold-start blind spot) (#271)

4134721 docs: archive DECOUPLED_COMPUTE ? DELIVERED (M0-M2 + M3.5 on GCP) (#270)

c8e60ee fix(gen0): provenance-stamp the manifest git_sha off a git-archive tarball (#269)

e6110ce feat(api): GET /api/health + return download_url from /api/compile (M1/P6) (#268)

a555887 docs(deploy): M2 FINALIZED ? Gen-0 weights/0.1.0-l4 pushed; both VMs deleted (\$0) (#267)

b05401b docs: drop DigitalOcean ? GCP is the sole compute stack (ADR-013) (#266)

4d546ec fix(gcp): teardown defaults to DELETE (a stopped VM still bills its disk) (#265)

dd30e9b feat(workspace): per-video workspace convention ? design doc + Phase 0 helper (#264)

dc9111b docs(adr): ADR-012 ? GPU-native compute; TPU deferred; Cloud Run+GPU is the serving model (#263)

d8c5aa0 docs(deploy): report polish ? highlight reel smoke-test result + VM-deleted final state (#262)

0d73d96 docs(deploy): first real GCP-L4 run report + gpu_setup conda-ToS/deps fix + Ops Agent runbook (#261)

380a4d0 docs(decoupled): narrow DECOUPLED_COMPUTE scope (M0?M2 + M3.5); defer M3/M4; D0=train, GCP=serve (ADR-011) (#260)

36ed29f feat(wasb): native runner defaults to streaming + correct M3.5 parity gate to a tolerance (#259)

38c2d0f feat(storage): gdrive:// reads a locally MOUNTED Google Drive (no API) (#258)

55ae408 docs(decoupled): GCP pivot runbook + D0?GCP ADR-010 (deploy/gcp + handoff) (#257)

dc53db5 feat(detectors): Linux-native WASB runner (M3.5) + worker real-trajectory extraction (#256)

5c77f48 docs(decoupled): HANDOFF.md ? session checkpoint / resume-here (#255)

d5407eb feat(worker): Gen-0 train-from-bucket (worker train) on a CPU stub-trajectory mock (#254)

814d813 feat(deploy): GPU machine provisioning ? NVMe scratch + native WASB setup (#253)

f06c6a6 fix: cross-platform canonicalization (Linux/macOS, not just windows) (#252)

6e5ac9e feat(worker): python -m backend.worker dispatcher ? bucket-driven train/infer (#251)

d292854 docs: SETUP_NEW_MACHINE ? canonical zero?train/infer runbook (#250)

6878d45 feat(storage): pluggable StorageBackend ? local + S3/GCS first-class buckets (#249)

4. user (2026-06-22T01:55:58.447Z)

Next steps (live) ? refreshed 2026-06-21

> **How to read:** this top section is the **current cross-track state + prioritized pending list** (refreshed 2026-06-21, replacing the stale 2026-06-13 header). The detailed **owned-model & rally-quality track** plan (2026-06-13) is **preserved below, still valid** ? it is now **one of several active tracks**. Live blow-by-blow: memory `current-state.md`. Doc lifecycle/hygiene: `docs/DOC_STATUS.md`.

? Where we are (2026-06-21)

DECOUPLED_COMPUTE delivered on GCP + archived (#270); **compute** = GCP only (ADR-013, D0 dropped). **COMPUTE_DECOUPLED_SERVING** subproject spun up (#272, **ADR-014**) ? one pure core + local/cloud profiles; **owner M0** design sign-off pending. **Per-video workspace P0** shipped (#264); **doc-status register + archival due-diligence** in place (#273). The **khelsutra** site + **per-rally UI** ship in parallel (sibling track). The **rally-quality / owned-model** track (below) is the core ML work, gated on **growing the golden corpus**.

? Prioritized pending ? all tracks

P0 ? highest leverage / unblocks the most

0. **owner-priority, 2026-06-21** MAKE IT UI-DRIVEN + ALPHA-SHAREABLE ? "a few recordings lying around ? highlights, from a page, no SSH". Today the cloud flow is **SSH/CLI-only** (`docs/CLOUD_GPU_DRYRUN_GUIDE.md`); **M12** does NOT fix it. Needs (a separate **khelsutra web/SPA** + deploy track, NOT an engine seam): **(i)** deploy the engine as a reachable **service** (M7 ? Cloud Run service, ? owner GCP spend for a **persistent** endpoint), **(ii)** the **SPA ingest/progress screen** (**khelsutra web/ B-P2: enter/upload video ? /api/process ? poll /api/jobs ? reel; ZERO engine code**), **(iii)** flip **M9 edge-auth ON** (merged #290, default-OFF) to gate alpha tenants, **(iv)** a **compute PICKER** in the UI ("your machine vs our cloud", D13/D15). The engine seams make it **possible**; the SPA makes it **usable**. Full record: `[ui-driven-cloud-serving-gap]`.

0b. **packaging track, IN PROGRESS** ? 11 PRs merged 2026-06-21/22] Downloadable batteries-

included app ? `docs/PACKAGING\_PLAN.md`.** The LIGHT-tier engine is bundle-ready: `pip install`
? `indexer --app` serves the **bundled SPA** single-origin, **torch-free**, with app-home state
/ ffmpeg resolver / DNS-rebinding guard / single-instance lock / safe DB upgrades (P0a?e, P1a/b,
P2-launcher/P2a/P2d/P2e). **? Next (fresh focused pushes):** ***(1)** P3 **GPU validation +
screencast** (L4, ~\$0.50, DELETE-after, \$100 cap ? `[[gcp-vm-always-delete]]`; ? native-WASB
0-rally tuned-config blocker), ***(2)** P2b-install (uv/script + truly-local default config),
***(3)** P2c first-run wizard (cross-repo khelsutra SPA from `api/capabilities`), then P3-cloud
(Cloud Run) + P4 (ONNX/train/annotate = north star). Memory `[[packaging-app-plan]]` / `[[next-
session-packaging-app]]`.

1. ? **`COMPUTE\_DECOUPLED\_SERVING` M0?M9 LARGELY SHIPPED** (2026-06-21, ADR-014 ratified). M1?M4
(#279/#280/#282/#283), **M5 #286, M6 #287, M8 #288, M9-auth #290** all merged (default-OFF); the
real M7 L4 run validated the worker (22 rallies, VM deleted, \$0). All \$8 gating Qs locked
(D7?D17). **Owner residual** (not blocking Claude's default-OFF code): counsel ToS/DPA
(M9-consent live + M11 live), real GCP prices+FX (M8 calibrate), Google OAuth app (M12 real), CF
team-domain/AUD + ingress-lock; real cloud verification authorized within a **\$100/?10k cap**
(D17).

2. **[owner, ongoing] Grow the golden corpus** ? multi-court badminton + ?3 matches/sport (TT
first). The single highest-leverage move; the quality track *and* full-corpus training feed on
it.

3. **[owner + Claude · GPU] Full-corpus training (n?5)** ? a real `weights/<ver>/` (the
`0.1.0-l4` bundle is n=2 plumbing-only).

P1 ? near-term, mostly Claude-doable (default-OFF, green-gated)

4. **[Claude] `COMPUTE\_DECOUPLED\_SERVING` batch ? REMAINING: M11 + M12** (M5/M6/M8/M9-auth ?
merged this session). **M11** = `backend/flywheel.py` capture?store?shadow-eval?goldenize
plumbing reusing `workspace.py::for\_video` + `backend/eval/experiment.compare` +
`backend/eval/promotion.py::promote\_if\_served\_safe`, behind `flywheel.\*`=OFF + a faked
`consent\_attested` (? privacy-sensitive ? give it fresh focus). **M12** =
`backend/storage/gdrive\_api.py` `GDriveApiStorage(StorageBackend)` + a `gdrive-api://` scheme
(regex already allows the hyphen; add to `\_KNOWN` + `\_backend\_class` + the StorageConfig
Literal) against an INJECTED fake Drive (mirror `gcs.py` DI); keep the mount-only `gdrive://`
untouched. ? **M9-consent cols = a new v7 migration** (M8 took v6). M9-consent ToS/DPA text =
owner/counsel.

5. **[Claude] Per-video workspace P1?P6** (P1 v6 DB migration ? P2 worker ? P3 per-video routing
of `detector/`+`tmp/` *`[the `output/chunks_` de-hardcode base = ? done via #282]\*` ? P4 ? P5
Launch-Report flip ? P6 GPU parity). ? P1's v6 migration coordinates with the M8 v6 cost-ledger
migration ? author the version bump once.

6. ? **Ops-Agent codified (#291)** ? `deploy/gcp/create\_gpu\_vm.sh` (the one-command VM helper)
always bakes the Ops-Agent startup-script; provision.sh grants the APIs+SA roles. PROVEN active
on a fresh VM 2026-06-21. ([[gcp-ops-agent-enable]])

7. **[owner] Set the ship-gate target (#172)** ? needs corpus growth.

8. **[owner-side GPU] R-series: promote-or-reject `R2` to `F1` as the gate** (the ablation) ?
needs GPU-extracted golden features.

P2 ? later / gated / deferred

9. Rally-quality **P3/P4** (person-fusion LOVO, audio cue) ? needs GPU golden features.

10. **M0.4 scorer swap** ? TemporalMaxer (per the 2026-06-17 external review).

11. **Multisport** (table-tennis first, then pickleball).

12. Serving **M8/M9** ? per-video cost ledger, edge auth, DPA/consent (within
`COMPUTE\_DECOUPLED\_SERVING`; owner/legal-gated).

13. **WASB-C3** weight provenance (gates *sharing* a trained model) ? legal.

14. **Rename repo off "badminton"*** (owner, BACKLOG).

15. Owner housekeeping: retire the shared **DO droplet** + **DO token**; collector ***md5***
export (collector repo).

16. **PMF field validation** (owner-side, business track).

Doc-hygiene (tracked in `docs/DOC\_STATUS.md` §4): ? #1 per-video reconcile + ? #2 this
refresh; **pending:** field/PMF/Roora ? archives + a data-layer `docs/` dir (owner boundary
confirm); confirm the COMPLETED top-level stubs; verify `HOSTING\_PLAN`/`UI\_PROPOSAL` currency.

Owned-model & rally-quality track ? detail (2026-06-13 plan; still valid)

****Updated:**** 2026-06-13 (****rally-detection quality track****: fusion + experiment-harness design docs merged,
owner-gated ? see the dedicated section below, the desktop-agent pick-up). Prior: 2026-06-10 (audit?M0?Gemini-battery session; later same day the ****UI-alpha track opened****:
ADR-009 ratified ? KhelSutra Guru / khelsutra.guru, private `khelsutra` repo, local-first delivery ? see
`docs/UI\_ALPHA\_EXECUTION\_PLAN.md`; ****PR-1 async jobs (#16) MERGED 2026-06-11, PR #87****; ****PR-2 upload/download MERGED 2026-06-11, PR #99**** ? live-E2E-verified (3-part chunked upload MD5 byte-identity +
compile?browser-download); next: PR-3 identity. Same day: the ****8?9 quality sweep #90?#98 landed**** ? ruff/mypy/
coverage-floor CI lanes, SQLite conn-close fix, config unknown-key warning, hybrid-twin collapse into
`trajectory\_hybrid.py`, WSL tilde+abspath live-path fixes; suite 388 green; LOVO 0.626 re-verified exact).
****Authoritative roadmap:****
`docs/OWNED\_MODEL\_IMPLEMENTATION\_PLAN.md` (+ ADR-008 for topology). ****Latest blow-by-blow:****
memory
`current-state.md` / `session-handoff.md`. This file = the prioritized "what to do next,"
refreshed each session.

? Next-session / GPU-box pickup (2026-06-15)

****M0 in-container quick wins are COMPLETE:**** ship-gate provisional blocker (****#170****),
train/eval split guard (****#171****), the ****doc-consistency sweep**** (****#174**** ? corrected 16
drifted docs vs the code), and the ****P3 person-feature fusion litmus machinery is already present + suite-green**** (``backend/eval/fusion_compare.py`` + ``ablation.py``: person-driven ?F1 /
bootstrap CI / paired sign test; the real-golden `skipif` litmus waits only on GPU-extracted data). ? CI is red repo-wide due to a ****GitHub Actions minutes/billing outage**** (account-level,
not code) ? work is local-gated + ``--admin``-merged until Actions is restored (Settings ? Billing
? Actions minutes / spending limit).

****Immediate next step ? the real P3 LOVO measurement**** (this box has a GPU + ``torch+cu124`` +
``cv2``; videos and trajectory CSVs are present, but the GPU-extracted golden features +
``.rallies.csv`` labels are not yet here):

1. ****[owner / GPU box]**** ``python -m backend.eval.fusion_golden --manifest output/human_lovo_manifest.json --out-dir "%USERPROFILE%\OneDrive\rally-annotator\golden-data\features\2026-06-15-n6"`` ? produces ``*_golden_features.json`` for the 6 golden videos. (GPU + WSL + videos + trajectory CSVs + ``.rallies.csv`` labels are owner-side only.)
2. Artifacts land in the ****shared cloud folder**** ``C:\Users\avidu\OneDrive\rally-annotator\golden-data\`` (OneDrive auto-syncs both ways). Convention + workflow:
****[`GOLDEN\_DATA\_SHARING.md`](data_pipeline/GOLDEN_DATA_SHARING.md)****; tracked in ****#175****.
3. ****[CPU dev/agent session]**** ``python -m backend.eval.fusion_compare --features-dir <shared>\features\2026-06-15-n6 --record`` + ``python -m backend.eval.ablation --features-dir <shared>\... --granularity group`` ? real person-feature ?F1; copy the JSONs into ``output/`` to also pass the ``skipif`` litmus tests (``tests/test_ablation.py::test_litmus_reproduces_gate_a``).

****Also queued (owner-gated / pending input):****

- Ship-gate target calibration ? ****#172**** (owner decision; needs corpus growth).
- Tests reorganization ? blocked on the owner's grouping metric (provide it ? an agent will do it).
- Restore CI ? the Actions outage is account-level (billing/minutes), not a repo/config bug.

****Leaving (bulky / M2 ? no start without owner go):**** ActionFormer (M0.4), event-index DB migration (M0.1), Gemini-silver purge (M0.3), cost-to-Gen-2 benchmark + ByteTrack deprecation, multisport, repo rename, PMF execution.

Where we are (one paragraph)

ADR-008 ratified ("repo split driven by productionization, not isolation"): training lives in-repo (``training/``)
behind a CI-enforced import wall; the featurizer is single-sourced
(``backend/features/rally_seq.py``, `train==serve`);
the ****Gen-0 harness ships**** (``python -m training.gen0.harness --manifest output/human_lovo_manifest.json --floor eval_baselines/heuristic_lovo_n6.json --version <semver>``) and produced artifact v0.1.0 (LOVO

****0.626****, floor passed,
sign test 5/6 wins $p=0.109$? honestly not significant, ship=False). The ****n=6** owned corpus is in the collector**
(262 rallies, Proprietary, commercial export = owned data only after the ShuttleSet reclassification). The ****Gemini** battery is done** (see table) ? the moat is quantified.

The quality table that now drives strategy (IoU 0.5 vs human golden labels)

Path Mahadevpura (clean-ish) Kushagra (multi-court)
--- --- ---
Heuristic (velocity windowing) 0.657 0.157
Gen-0 owned head (LOVO, n=6) 0.744 0.561
Two-stage WASB+Gemini refine 0.83 ?
Gemini wrapper (`gemini-flash-latest`) **0.93** **0.66**

****Reading:**** clean footage is commoditized (serve it with Gemini for cents). ****Multi-court** drops the commodity wrapper to 0.66 ? that is the owned model's ship target** (its FPs are background-game pickups + dead-time merges, the exact contrast-metric confound). Gen-0 is 0.10 behind with only n=6 ? a corpus-growth-sized gap, not an architecture-sized one. Two-stage refine = the cost-efficient serving path for LONG tapes (uploads candidate clips only); wrapper 503-flakiness (observed all session) makes the provider-fallback registry load-bearing.
Baselines tracked in `eval\_baselines/` (heuristic_lovo_n6.json, gemini_wrapper_2026-06-10.json).

Rally-detection quality track (design merged 2026-06-13, owner-gated; implementation in progress)

****This is the track a desktop/local agent is slated to pick up.**** Design docs landed (PR #112 + session). All implementation is ****owner-gated**** (per explicit in the plan); each step = ONE PR off `master`.

****Hardening loop** (T1-T5 positive tests + audit + finale) COMPLETE (2026-06-14, PRs #156/#157/#160/#161/#163 + #165 + final records).** See archived `docs/archives/CODE\_AUDIT\_AND\_TEST\_HARDENING-COMPLETED-2026-06-14.md` (Living Log with every pre/post/review) and `docs/archives/HARDENING\_LOOP\_HANDOFF-COMPLETED-2026-06-14.md` (full rules + STATE). Top-level now pointer stubs per #165 review. All 5 items + verification + archive done; clean slate.

- ****Docs:**** `docs/MULTI\_SIGNAL\_FUSION\_PLAN.md` (shuttle + person + audio, one fusion layer) .
`docs/EXPERIMENT\_HARNESS.md` (A/B + shadow + promotion gate, zero-regression) .
`docs/ROORA\_LABELING\_GUIDELINES.md` (v8) . `docs/HOW\_RALLY\_DETECTION\_WORKS.md` (2-layer mental model).
- ****Key finding** (no re-investigation needed):** the owner's "held-shuttle counted as a rally" bug is ****NOT** a missing capability ? `backend/pipeline/detectors/rally\_gate.py` already implements the net-crossing ****"Gate A"*** (`apply\_gate(..., min\_crossings=2)`) that kills service-feed/held-shuttle windows, but it is ****only** called from eval drivers** (and there was no `min\_crossings` knob in `IndexingConfig`). ****Wiring Gate A** into the live segmenter (now via fusion_hybrid) was the single cheapest, highest-confidence quality win.** (P2 FusionSegmenter specifics: registry `fusion\_hybrid`, ****default-OFF****, seeds from substrate, persists `net\_crossings` + optional person features, optional served Gate A/B via config knobs; adversarially reviewed; suite green. See `tests/test\_served\_gate\_a\_regression.py` for the honest finding that on production windowing Gate A is neutral/negative in some cases, hence kept opt-in with knob for safety, and the leaderboard revert note.)
- ****Current status** (2026-06-14):** P1 (person cue extraction to reusable `PersonDetector`) + P2 (FusionSegmenter + Gate A signal + served handoff gating) + experiment harness P1 DONE (adversarial review + NO-REGRESSION; suite green). ****P3 next** (this PR + follow-ups):** learned fusion (person features ? harness `compare` LOVO lift on the 6 golden videos; eager-person-compute optimisation); then P4 audio via hit_detector.
- ****Pick-up order** (each = ONE PR off `master`, owner approval first, flag-gated default-OFF, promote only on measured LOVO):**
 1. (P2 first-step, largely done via fusion_hybrid) Wire Gate A into production served path + config knob (min_crossings); apply in hybrids at runtime. (Now available opt-in; served neutral on production windowing per litmus ? kept opt-in with knob for safety.)

2. P3 first step (this work item): harness `compare` LOVO litmus for person-on variant (players_in_court + two_sided + colocation etc. from `fusion\_features`) on the 6 golden; add the eager compute optimisation note/landing. Pure + harness (testable here).

3. P3/P4 continuation: audio cue integration + full learned promotion if lift proven.

- **Discipline** (from EXPERIMENT_HARNESS.md): every new cue **flag-gated**, default OFF;
promote to default **only** on measured LOVO lift through `run\_regression.py` (exit 0/1/3);
pinned **CONTROL** ? rollback = config flip. Harness re-uses ~80% existing code.

- ? Gate A, person cue logic, harness, and pure fusion paths are testable here; the heavier real-video / golden LOVO / ablation confirmation waits on the GPU-extracted golden features + **rallies.csv** labels, which are not yet present in this checkout.

- **External due-diligence review inputs** (2026-06-17) ? **PICK UP AFTER** current ideas are exhausted; full details + citations in [**RALLY_QUALITY_RESEARCH.md**](RALLY_QUALITY_RESEARCH.md) §5.6. **An owner-commissioned 31-reference review of the quality report** **validated** the methodology and surfaced these **future** levers (research inputs, not active work, not new claims; verify citations before any submission):

1. **"First-mile" paper framing** (highest-value): **stroke-classifiers** (BST/DSTA) + the big datasets (ShuttleSet ~36k strokes, Fine-Badminton ~27k actions) all **assume** a pre-trimmed rally clip already exists ? we solve the ignored prerequisite. Pitch the paper as **"the missing untrimmed-video preprocessing pipeline that lets BST/MonoTrack run on raw footage."**

2. **Ground R2 in Action-Spotting / Precise-Event-Spotting** (SoccerNet/TennisSet/T-DEED) ? turns R2 into a defensible publishable contribution.

3. **Add domain baselines for any submission:** MonoTrack + BST on ShuttleSet / Fine-Badminton (we have only heuristic + Gemini today).

4. **Prefer TemporalMaxer over ActionFormer** for the M0.4 scorer swap (parameter-free max-pool, ~2.8× fewer GMACs, small-N robust) ? see "M0.4 next increment" below.

5. **Court polygons ? pose ? the rally-END lever:** masking on-court players can resolve **"occluded end-of-rally states"** (the #1 remaining gap vs Gemini), not just spectator noise.

6. **Name the eval?serve lesson** "pipeline distribution skew" ? a citable paper "lesson."

Discipline reminder: owner approval before starting any owner-gated item; one PR per step; babysit the PR (poll/fix/reply) until merge observed, **then** advance.

Prioritized next steps

1. **[owner, in progress] GROW THE GOLDEN CORPUS** ? the single highest-leverage move; every lever below feeds on it.

Priority order for new videos: **(a) multi-court badminton** (the target distribution AND where the ship target

lives), **(b) ?3 matches per new sport ? table tennis first** (WASB transfers at F1 0.909; pickleball labels are

corpus-gold but not trainable until a clean MIT/Apache ball detector exists), (c) capture variety per

VIDEO_RECORDING_GUIDELINES.md; **adult sessions only for the training pool** (consent guardrail). Per video:

label ? **curate import-local --normalize --license Proprietary --fps <true fps>** ? re-run the Gen-0 harness.

The paired sign test needs n: at n=10 with 9 wins, p?0.011 ? significance is reachable this month.

2. **[owner decision] Set the ship-gate target** ? now informed: floor = heuristic 0.480 (necessary), **multi-court**

reference = wrapper 0.66 (the number that makes the owned model worth serving), clean-footage ceiling = 0.93

(not the target ? Gemini serves that tier). Suggested shape: "LOVO F1@tIoU0.5 ? 0.70 on multi-court folds with

paired-sign p<0.05 vs heuristic" ? owner to ratify/adjust.

3. **M0.4 next increment (\$0/local):** swap the Gen-0 LogReg for a lightweight temporal head (MIT, minutes on the

2070S ? NOT the heavyweight Gen-2 stack) inside the existing harness; per-video threshold calibration (the

LOVO-vs-oracle gap is visible per fold in the harness output). **Prefer TemporalMaxer** (parameter-free

max-pool, ~2.8× fewer GMACs, small-N robust) over ActionFormer/ASFormer per the 2026-06-17 external review

(**RALLY_QUALITY_RESEARCH.md** §5.6) ? better fit for the cheap/local/fast mandate.

4. **Multisport thread:** ingest Roora pickleball/TT CSVs (**import-local --normalize**); merged-

prototype LOVO across

badminton+TT once TT has 3 matches; identify a clean pickleball ball detector. ? TT-on-WASB serving stays gated

by C3 (provenance multiplies per sport).

5. **PMF validation (cheaper than any engineering month):** C13 academy interviews (+ the two added questions:

"multi-court footage nobody watches?" / "pay per-match to index it?"); camera pilot at the warmest 2-3 academies

(consent at capture; demo served via the Gemini path or human-polish ? the reel pattern exists:

`output/MahadevpuraSingles\_highlights.mp4`). **Field kit COMPLETE (2026-06-12, owner ratified the

physical-visit approach; reframed product-feedback-first per owner spec):**

`docs/PRODUCT\_FIELD\_ASSOCIATE\_FLYER.md` (the recruiting flyer ? owner fills ?/contact placeholders and recruits) +

`docs/FIELD\_VISIT\_PLAYBOOK.md` (the hired associate's visit playbook: coach + student script, demo-after-Q6

rule, ground rules, Phase-2 record-and-show-back stretch goal) +

`docs/FIELD\_VISIT\_ANSWER\_SHEET.md` (per-visit

capture form incl. demo-reaction + student blocks) + `docs/PMF\_FIELD\_SIGNALS.md` (G/A/R anchors + **pre-registered

decision rules** ? PROCEED / PIVOT A-C / FALSIFIED thresholds fixed before the first visit).

6. **Housekeeping:** ~rotate the Gemini API key~ **DONE 2026-06-10** (owner-verified working; confirm old-key

deletion in AI Studio); **rename repo off "badminton" (owner action,

in BACKLOG); collector=SoR confirmation; off-peak niceties: none pending ? the battery completed.

Pointers / artifacts

- Roadmap: `docs/OWNED\_MODEL\_IMPLEMENTATION\_PLAN.md` · Topology: ADR-008

(`docs/archives/decisions/DECISIONS.md`) ·

Study: `docs/archives/research/OWNED\_MODEL\_TRAINING\_STUDY.md` · Quality history:

`docs/archives/quality-iterations/2026-06-multivideo-lovo/`.

- Harness: `training/gen0/harness.py` (one command ? train ? LOVO ? gate ? artifact bundle; weights gitignored,

manifest reviewable). Featurizer: `backend/features/rally\_seq.py` (FEATURE_SCHEMA v1).

- Corpus: collector DB (system-of-record, 6 matches/262 rallies) · eval manifest

`output/human\_lovo\_manifest.json`

(gitignored) · durable trajectories + Roora files: `C:\Users\avidu\Projects\Annotation Setup\Collect\Trajectories\` ·

labels: `...\Collect\Golden Labelled\`.

- Gemini ops learnings: serial uploads (`indexing.ai\_max\_workers=1`) + chunk re-encode

(`indexing.ai\_chunk\_scale\_width=1280`) + 6-attempt exponential generate retry ? a cold/prepaid project burst-throttles

with a MISLEADING "credits depleted" 429; ramp gradually.

- ? No owned-model implementation beyond greenlit M0.x until owner greenlights the next milestone. Committed next

PLATFORM slice = async job model (single-tenant).

5. assistant (2026-06-22T01:56:02.994Z)

The NEXT_STEPS doc has detailed claims. Now let me verify the key quality-track claims against the actual code on origin/master: did Gate-A wiring, FusionSegmenter, R2-metric, R4-end-rule, the #146/#151 flip actually ship? Let me read the other docs and grep the code.

6. user (2026-06-22T01:56:07.312Z)

=== rally_gate.py exists on master? ===

100644 blob 626906513b8adb0bde9d9110032f36014d5161ed backend/pipeline/detectors/rally_gate.py

=== fusion segmenter registered? ===

origin/master:backend/pipeline/segmenters/__init__.py:7:from

backend.pipeline.segmenters.fusion_hybrid import FusionSegmenter

origin/master:backend/pipeline/segmenters/__init__.py:17: "FusionSegmenter",

origin/master:backend/pipeline/segmenters/fusion_hybrid.py:23:nothing is gated (thresholds 0,


```

person OFF), so `FusionSegmenter` with default config seeds
origin/master:backend/pipeline/segmenters/fusion_hybrid.py:24:and hands off identically to
`wasb_hybrid`. Registered as ``fusion_hybrid``, NOT the default
origin/master:backend/pipeline/segmenters/fusion_hybrid.py:39:class
FusionSegmenter(TrajectoryHybridSegmenter):
origin/master:backend/pipeline/segmenters/fusion_hybrid.py:56:         return "fusion_hybrid"
origin/master:backend/pipeline/segmenters/fusion_hybrid.py:189:segmenter_registry.register(Fusio
nSegmenter)
origin/master:backend/pipeline/segmenters/trajectory_hybrid.py:119:         rows;
``FusionSegmenter`` overrides this to add ``net_crossings`` (+ person-cue
origin/master:backend/pipeline/segmenters/trajectory_hybrid.py:133:         ``FusionSegmenter``
overrides this to apply Gate A/B when thresholds are raised.
origin/master:backend/pipeline/segmenters/trajectory_hybrid.py:158:         # FusionSegmenter the
windows are still Gate-A/B-filtered via `_select_for_handoff`.
=== min_crossings in IndexingConfig? ===
origin/master:backend/config/models.py:100:         threshold is raised
(``min_crossings``/``min_players`` default 0 = OFF). The court mask
origin/master:backend/config/models.py:115:         min_crossings: int = Field(default=0, ge=0)
=== Gate A served regression test? ===
100644 blob 27779ff2fb94e0ed2a3ae510c61e2c91931f6f02    tests/test_served_gate_a_regression.py
=== run_regression exit codes ===

```

7. user (2026-06-22T01:56:09.605Z)

```

=== commits mentioning fusion/gate/R2/R4/146/151 ===
a240221 fix(test): ablation litmus uses canonical experiment._is_fusion_schema (unbreaks the
suite) (#237)
d749fb9 test(eval): re-ground served Gate-A litmus on the present (multicourt) golden set (#236)
ab03043 refactor(eval): single source of truth for the _is_fusion_schema predicate (#215)
67b6927 feat(config): R1 milestone ? flip windowing defaults ON (cap=6 + R4), re-verified at
n=13 (#204)
f5f339a fix(eval): fusion loaders skip windowing-only files on a mixed golden corpus (#210)
8c68143 feat(eval): R1 ? net over-seg promotion guard unblocks the cap6+R4 milestone (#201)
34b9caa feat(eval): R4 ? rally-END inactivity trim (default-OFF) + R3 cap re-scope finding
(#200)
7e8f88f feat(eval): R2 ? task-faithful tolerance-match rally metric (augments tIoU) (#199)
cf8973e docs(eval): R2 design ? task-faithful tolerance-match rally metric (FOR REVIEW) (#198)
950106b fix(fusion): W2 net-crossings gate is do-no-harm ? never empty a non-empty video (#188
follow-up) (#190)
428ee44 address follow-up review on #169: correct factual error in item-3 investigation note
(fusion_compare.py, fusion_golden.py, ablation.py + tests *do* exist per ls/grep on the tree;
person features in fusion_hybrid + detectors/fusion_features.py; remaining M0 in-container task
is wiring into compare). Cleaned duplicate sequencing paragraph. Verified files present. Plan
now accurate. M0 item 3 placement remains correct and even stronger.
b649f1c address review on #169: revert frozen archive edit (should-fix); trim handoff to short
pointer per #165 intent (record moved to plan); add Transition Record section; unbundle catalog
items 3/6/10 + clarify P3 'shipped' state with investigation (no fusion_compare.py etc. present;
general harness + P2 are); add M0 Quick Wins (the 4 in-container items per course-correction);
re-sequence milestones (M0 quick wins first + owner-led parallel, heavy last); update risk table
+ sequencing text. Thorough point-by-point in PR comment.
5fabffd address review for #166: retitled PR (via update); restored Gate-A cheapest-win
rationale, P2 FusionSegmenter specifics, served regression test finding/revert note in
NEXT_STEPS; removed long P3 prose comment from fusion_hybrid.py source (moved essence to docs);
confirmed archived snapshot Post-merge #165 addition is one-time finalization (no further edits
to frozen archive). Fixes per owner review points. Ready for re-review.
5cc0ea4 test + docs: hardening loop completion record + rally quality track P3 first step
(harness LOVO for person features); identify next candidates (per NEXT_STEPS). Small landing
comment in fusion_hybrid + explicit list in NEXT_STEPS. Syntax green. Pre: container baseline
(ast + obsreport note). Branched, one item, will babysit. Verification: owner re-run full + real
video.
a03f0f4 docs(quality): reconcile served-side Gate-A entry with the #151 revert (#153)
53bc743 docs(quality): served-side Gate-A check ? flip's offline win doesn't transfer to
production windowing (#152)
2cd469e revert(fusion): undo the Gate-A served-default flip (#146) ? confirmed served regression
(eval?serve) (#151)

```

```
bd2a216 feat(eval): served Gate-A verification + repeatable regression litmus (#146 follow-up)
(#148)
570cf2b feat(fusion): flip Gate-A (held-shuttle guard) to the served default ? config-reversible
(#146)
77082f4 feat(eval): wire run-telemetry black box into fusion_golden extractor (#140)
ec5bbe0 chore(eval): track experiment_leaderboard.json (fusion P3/P4 honest LOVO records) (#139)
74c1303 docs(quality): log fusion P3+P4 measured ?F1 on the n=6 golden corpus (#137)
```

=== #146 and #151 in full history ===

```
a03f0f4 docs(quality): reconcile served-side Gate-A entry with the #151 revert (#153)
2cd469e revert(fusion): undo the Gate-A served-default flip (#146) ? confirmed served regression
(eval?serve) (#151)
bd2a216 feat(eval): served Gate-A verification + repeatable regression litmus (#146 follow-up)
(#148)
570cf2b feat(fusion): flip Gate-A (held-shuttle guard) to the served default ? config-reversible
(#146)
```

8. assistant (2026-06-22T01:56:13.298Z)

This is very revealing. The history shows a lot has shipped since the NEXT_STEPS doc's detailed section was written. Critically, R1/R2/R4 all merged, and there's an n=13 re-verification. Let me dig into the R-series and the current windowing defaults state, plus check what the most recent quality work is.

9. user (2026-06-22T01:56:17.418Z)

```
=== R1 milestone: windowing defaults ON (#204) ? verify config defaults on master ===
origin/master:backend/config/models.py:104: # Which trajectory substrate seeds the windows
(shuttle stays primary).
origin/master:backend/config/models.py:113: # Served Gate A: drop windows whose shuttle net-
crossings < this from the AI handoff.
origin/master:backend/config/models.py:116: # Served Gate B: when the person cue is on, drop
windows with median in-court players
origin/master:backend/config/models.py:147: min_action_window_duration: float = 2.0
origin/master:backend/config/models.py:148: # BOUNDED WINDOWING (over-merge fix W1,
RALLY_QUALITY_RESEARCH.md §6). 0 = OFF; > 0 = a window
origin/master:backend/config/models.py:149: # longer than this many seconds is recursively
split at its largest internal inactivity gap (?
origin/master:backend/config/models.py:150: # ``window_min_split_gap`` s ? the most likely
rally boundary), so one window can't swallow
origin/master:backend/config/models.py:152: # *** R1 MILESTONE DEFAULT-ON (cap=6): the
smallest safe local-windowing flip. Measured n=13
origin/master:backend/config/models.py:153: # golden, R2 tolF1: baseline?milestone (cap=6 +
R4 below) ?+0.222, sign 12/0, p=0.0005; cap=6
origin/master:backend/config/models.py:154: # beats cap=12 ?+0.110, sign 12/0; net over-seg
guard PASS (mean merge_rate 0.651?0.161). ***
origin/master:backend/config/models.py:155: max_window_duration: float = 6.0
origin/master:backend/config/models.py:156: window_min_split_gap: float = 0.5 # min internal
gap (s) that qualifies as a split point
origin/master:backend/config/models.py:157: # HARD-CAP fallback for W1: when True, an over-
long window with NO internal inactivity gap
origin/master:backend/config/models.py:159: # chopped at its midpoint until ?
max_window_duration, making the cap a true upper bound.
origin/master:backend/config/models.py:161: window_hard_cap: bool = False
origin/master:backend/config/models.py:162: # R4 rally-END inactivity trim (s, 0 = OFF).
After a rally the shuttle keeps moving slowly
origin/master:backend/config/models.py:163: # (picked up / walked) above velocity_thresh, so
the window over-extends its END (the measured
origin/master:backend/config/models.py:165: # window stays dense with FAST motion (velocity >
inactivity_rally_thresh). Only cuts.
origin/master:backend/config/models.py:166: # *** R1 MILESTONE DEFAULT-ON (1.5/0.20/0.30): on
top of cap=6, R4 adds a small but significant
origin/master:backend/config/models.py:167: # end-precision gain ? ?tolF1 +0.040, sign 9/1/3,
p=0.022, |?end| 4.96?3.31s (n=13). The W1 cap
origin/master:backend/config/models.py:168: # is the dominant lever; R4 is the marginal
```

```

increment. See QUALITY_ITERATIONS.md "R1". ***
origin/master:backend/config/models.py:169:    window_inactivity_end: float = 1.5
origin/master:backend/config/models.py:170:    inactivity_rally_thresh: float = 0.20
origin/master:backend/config/models.py:171:    inactivity_min_density: float = 0.30
origin/master:backend/config/models.py:173:    # the W1 gap-split AND the R4 inactivity trim
have each had their chance, a group STILL longer
origin/master:backend/config/models.py:174:    # than window_blob_factor × max_window_duration
(no internal gap, no rally-end signal ? e.g.
origin/master:backend/config/models.py:175:    # mahadevpura-1's ~65s residual ? 11× a 6s cap)
is midpoint-chopped to ? the cap as a FORCED cut
origin/master:backend/config/models.py:176:    # (logged + flagged on the window) so the
degenerate single-window outcome is avoided and the
origin/master:backend/config/models.py:177:    # clip is surfaced as needing rally-STATE cues,
not a bigger cap. Differs from window_hard_cap
origin/master:backend/config/models.py:178:    # (which chops BEFORE R4 and over-segments normal
clips): blob-fallback runs AFTER R4 and only

```

=== full quality-track commit list (broader net) ===

```

36ed29f feat(wasb): native runner defaults to streaming + correct M3.5 parity gate to a
tolerance (#259)
e358b28 feat(eval): prominent-court window gate ? drop multicourt background-court rallies
(default-OFF) (#243)
20f8aa3 feat(eval): P7 long-rally (>5s) quality lens + reel knob (both default-OFF) (#220)
a010667 feat(eval): nightly all-OFF floor ? pin a baseline_off reference config (#217)
67b6927 feat(config): R1 milestone ? flip windowing defaults ON (cap=6 + R4), re-verified at
n=13 (#204)
f5f339a fix(eval): fusion loaders skip windowing-only files on a mixed golden corpus (#210)
be39a70 docs: rally-detection quality report ? technical-paper checkpoint (2026-06-17) (#206)
4bbd987 docs: rally-detection quality report ? technical-paper checkpoint (2026-06-17)
f5f5e52 feat(eval): R5 ? blob-fallback splits the continuously-active blob (default-OFF) (#203)
8c68143 feat(eval): R1 ? net over-seg promotion guard unblocks the cap6+R4 milestone (#201)
2eaccd5 feat(eval): nightly rally-quality regression tripwire (golden corpus, CPU) (#202)
34b9caa feat(eval): R4 ? rally-END inactivity trim (default-OFF) + R3 cap re-scope finding
(#200)
7e8f88f feat(eval): R2 ? task-faithful tolerance-match rally metric (augments tIoU) (#199)
cf8973e docs(eval): R2 design ? task-faithful tolerance-match rally metric (FOR REVIEW) (#198)
c483346 docs(quality): real-footage validation capstone + first ground-truth at-par + dual-eval-
set discipline (#196)
bb4d6e2 feat(eval): Tier 1 (W1.5) ? hard-cap fallback enforces the window cap on continuously-
active trajectories (#195)
8d3e185 feat(eval): golden fixtures P1 ? de-attribution/minimization tool + M2 refusal gate
(#192)
64494ef feat(eval): golden regression fixtures M1 ? video-free CONTROL-only characterization
harness (#189)
a471745 feat(eval): Tier 1 (W2) ? net-crossings rally-state gate on W1 (?F1 +0.019, 6/0) (#188)
a3b209a feat(windowing): Tier 1 (W1) ? bounded windowing fixes the over-merge (?F1 +0.139, 6/0)
(#187)
90ab087 feat(eval): Tier 0 (E1) ? over-segmentation-aware rally-seg eval + frozen baseline
(#185)
960c00d docs: rally-quality research + eval-framework roadmap (foundation doc) (#184)
c558372 Merge pull request #166 from avidullu/test/rally-quality-p3-harness
5cc0ea4 test + docs: hardening loop completion record + rally quality track P3 first step
(harness LOVO for person features); identify next candidates (per NEXT_STEPS). Small landing
comment in fusion_hybrid + explicit list in NEXT_STEPS. Syntax green. Pre: container baseline
(ast + obsreport note). Branched, one item, will babysit. Verification: owner re-run full + real
video.
4251c9c fix(eval): close the eval?serve windowing gap ? single-source SERVED preset + served-
regression promotion guard (#154)
a03f0f4 docs(quality): reconcile served-side Gate-A entry with the #151 revert (#153)
53bc743 docs(quality): served-side Gate-A check ? flip's offline win doesn't transfer to
production windowing (#152)
bd2a216 feat(eval): served Gate-A verification + repeatable regression litmus (#146 follow-up)
(#148)
e75ec35 feat(eval): LOS0/LOG0 ablation runner ? per-signal marginal ?F1 (Phase 0) (#147)
570cf2b feat(fusion): flip Gate-A (held-shuttle guard) to the served default ? config-reversible

```

```
(#146)
7c6498f docs(quality): personalization Phase-0 checkpoint ? eval/safety rails + locked
structural-guard decision (#145)
1a98ee8 feat(eval): per-user no-regression guard ? never make a user worse on a retrain/upgrade
(#143)
1b5af67 feat(eval): held-out-user learning-curve eval ? "<K videos to superior quality" as a
release gate (#142)
7376415 feat(eval): per-user promotion gate for flag-gated personalization (#141)
77082f4 feat(eval): wire run-telemetry black box into fusion_golden extractor (#140)
74c1303 docs(quality): log fusion P3+P4 measured ?F1 on the n=6 golden corpus (#137)
2e5892e feat(eval): fix the learned variant ? train-fold threshold tuning + calibration (C2)
(#130)
f680b8d feat(eval): wire C1+C4 honest reporting into experiment.compare() (additive) (#129)
d7b1f51 feat(eval): honest ?F1 ? bootstrap CI + paired sign test on the fusion lift (#127)
813d1f8 feat(eval): fusion_golden --smallest-first (quick-feedback extraction order) (#126)
```

10. user (2026-06-22T01:56:18.664Z)

=== R2 metric design doc + R-series in HOW_RALLY / FUSION_PLAN ===

How rally detection works (the 2-layer mental model)

A short conceptual map of how we go from raw video to rally `[start, end]` segments ?
and, crucially, **which layer our annotation can improve.**

```
...
video ??? frames ??? [DETECTION] ??? shuttle (x,y,visible) per frame
                        ?                               ?
                        (WASB CNN,                       per-frame trajectory
                        frozen weights)                  ?
                                                         ?
                                                         velocity signal (how fast the shuttle moves)
                                                         ?
                                                         [SEGMENTATION] ??? the 3 knobs we tune
                                                         ?
                                                         ?
rally windows [start, end] ??? IoU vs golden labels
...
```

Two layers ? keep them separate in your head

****1. DETECTION ? "where is the shuttle, this frame?"****

WASB is a CNN that, for each frame, outputs a heatmap ? the shuttle's `(x, y)` + a visibility/confidence. Its weights were learned (by gradient descent, on shuttle-coordinate labels) and we use them ****frozen****. We chose a ***shuttle*** detector on purpose: "shuttle in flight" is the signal for "a rally is happening," and it's ****camera-agnostic**** ? unlike player-motion, which fooled the old heuristic on broadcast cameras that pan with the players. (Your videos are static tripod, so player-motion would also work ? but shuttle-motion is the robust general bet.)

****2. SEGMENTATION ? "when is a rally?"****

From the per-frame `(x, y)` we compute the shuttle's ****velocity**** (displacement between frames). Then three knobs turn that into rally intervals:

- ****`velocity\_thresh`**** ? a frame is "active" only if the shuttle is moving faster than this (in flight, not lying on the floor / being picked up).
- ****`merge\_gap`**** ? active frames within this many seconds get joined into one rally (bridges brief gaps: a shuttle momentarily occluded, or the beat between two hits).
- ****`min\_window\_duration`**** ? throw away windows shorter than this (blips / false flickers).

Output: a list of `[start, end]` windows = the rallies.

Where the annotation fits (the key point)

Our Roora/golden labels are ****rally `[start, end]` times**** ? they say ***when*** rallies are, not ***where*** the shuttle is. So they can't teach WASB's detector (that needs `(x,y)` labels). They

****can**** grade and tune the ****segmentation layer****: we score predicted windows against the golden rallies with ****temporal IoU**** (how much they overlap), and pick the 3 knobs that maximise it ?
****calibrate on some rallies, measure on held-out ones**** so the number is honest.

Where sparks could land (open questions ? not commitments)

Things worth a shower-thought; any of these could beat 3 hand-tuned thresholds:

- ****Richer features than raw velocity:**** acceleration, trajectory **continuity** (a real rally is a smooth arc that survives occlusion), shuttle **height/arc**, or ****direction reversals**** (each reversal ? a shot ? ties to ``shots_count``!).
- ****A tiny learned segmenter**** instead of 3 thresholds: feed the trajectory features to a small model trained on the golden rally labels (still the segmentation layer, just smarter).
- ****Hysteresis**** (two thresholds: enter-rally high, stay-in-rally low) ? robust to mid-rally dips.
- ****Rhythm/periodicity:**** rallies have a hit cadence; dead time doesn't.
- ****Fusion:**** combine shuttle-motion (WASB) with player-motion for static cams. ? designed in MULTI_SIGNAL_FUSION_PLAN.md (one rally-fusion layer; the person cue is ADR-007's parked #2 and ``rally_gate.py``'s "Gate B" ? now built (default-OFF, unmeasured on golden) in ``fusion_hybrid.py``).

11. assistant (2026-06-22T01:56:24.456Z)

Now I have a clear picture. The R-series (R1/R2/R4) has shipped substantially. Key finding: ****R1** milestone already flipped windowing defaults ON (cap=6 + R4) and was re-verified at n=13** with strong significance (#204). This is a MAJOR correction vs the stale detailed NEXT_STEPS section which still lists "Wire Gate A" / "P3 person fusion" as next. Let me verify the current default state more precisely and check what's genuinely still open ? R5 blob-fallback (default-OFF), prominent-court gate (#243), and the person/audio fusion.

12. user (2026-06-22T01:56:28.684Z)

=== What is the LATEST quality commit on master? (timeline) ===

2026-06-19 21:43:58 +0530 e358b28 feat(eval): prominent-court window gate ? drop multicourt background-court rallies (default-OFF) (#243)
2026-06-19 21:40:32 +0530 626061d docs(rally): prominent-court detection design ? kill multicourt background-court FPs (#241)
2026-06-19 12:52:51 +0530 20f8aa3 feat(eval): P7 long-rally (>5s) quality lens + reel knob (both default-OFF) (#220)
2026-06-18 13:09:56 +0530 a010667 feat(eval): nightly all-OFF floor ? pin a baseline_off reference config (#217)
2026-06-18 11:23:56 +0530 67b6927 feat(config): R1 milestone ? flip windowing defaults ON (cap=6 + R4), re-verified at n=13 (#204)
2026-06-17 17:34:21 +0530 f5f5e52 feat(eval): R5 ? blob-fallback splits the continuously-active blob (default-OFF) (#203)
2026-06-17 17:24:48 +0530 8c68143 feat(eval): R1 ? net over-seg promotion guard unblocks the cap6+R4 milestone (#201)
2026-06-17 17:19:49 +0530 2eaccd5 feat(eval): nightly rally-quality regression tripwire (golden corpus, CPU) (#202)
2026-06-17 15:25:17 +0530 34b9caa feat(eval): R4 ? rally-END inactivity trim (default-OFF) + R3 cap re-scope finding (#200)
2026-06-17 14:54:21 +0530 7e8f88f feat(eval): R2 ? task-faithful tolerance-match rally metric (augments tIoU) (#199)
2026-06-16 22:54:45 +0530 c483346 docs(quality): real-footage validation capstone + first ground-truth at-par + dual-eval-set discipline (#196)
2026-06-16 18:21:22 +0530 bb4d6e2 feat(eval): Tier 1 (W1.5) ? hard-cap fallback enforces the window cap on continuously-active trajectories (#195)
2026-06-16 11:51:00 +0530 8d3e185 feat(eval): golden fixtures P1 ? de-attribution/minimization tool + M2 refusal gate (#192)
2026-06-16 10:34:10 +0530 64494ef feat(eval): golden regression fixtures M1 ? video-free CONTROL-only characterization harness (#189)
2026-06-16 10:08:21 +0530 a471745 feat(eval): Tier 1 (W2) ? net-crossings rally-state gate on W1 (?F1 +0.019, 6/0) (#188)
2026-06-16 09:49:16 +0530 a3b209a feat(windowing): Tier 1 (W1) ? bounded windowing fixes the

```

over-merge (?F1 +0.139, 6/0) (#187)
2026-06-16 09:30:46 +0530 90ab087 feat(eval): Tier 0 (E1) ? over-segmentation-aware rally-seg
eval + frozen baseline (#185)
2026-06-14 16:35:08 +0530 a03f0f4 docs(quality): reconcile served-side Gate-A entry with the
#151 revert (#153)
2026-06-14 16:28:42 +0530 53bc743 docs(quality): served-side Gate-A check ? flip's offline win
doesn't transfer to production windowing (#152)
2026-06-14 16:26:17 +0530 bd2a216 feat(eval): served Gate-A verification + repeatable regression
litmus (#146 follow-up) (#148)

```

```

=== R5 blob fallback + prominent-court gate: still default-OFF? ===
origin/master:backend/config/models.py:174:    # than window_blob_factor × max_window_duration
(no internal gap, no rally-end signal ? e.g.
origin/master:backend/config/models.py:185:    window_blob_factor: float = 4.0

=== person cue / Gate B default state ===
origin/master:backend/config/models.py:98:    shuttle net-crossing count (Gate A signal) and ?
only when ``cue_flags['person']`` is
origin/master:backend/config/models.py:99:    true ? the person-cue features, but it does NOT
gate the served handoff unless a
origin/master:backend/config/models.py:100:    threshold is raised
(``min_crossings``/``min_players`` default 0 = OFF). The court mask
origin/master:backend/config/models.py:102:    supplied ? off-court persons (spectators /
adjacent courts) are excluded.
origin/master:backend/config/models.py:110:    # Per-cue enable flags, e.g. {"person": true}.
Person cue is OFF unless explicitly
origin/master:backend/config/models.py:116:    # Served Gate B: when the person cue is on, drop
windows with median in-court players
origin/master:backend/config/models.py:118:    min_players: int = Field(default=0, ge=0)
origin/master:backend/config/models.py:121:    # Net line for the person two-sided-motion split
(person features only ? the shuttle

```

13. user (2026-06-22T01:56:31.553Z)

```

=== run_regression.py exit-code contract (the promotion gate) ===

=== QUALITY_ITERATIONS.md ? latest R-entry / current frontier ===
25:## Baselines that drive strategy (IoU 0.5 vs human golden labels)
41:## Current track (live) ? newest first
43:### P7 ? Long-rally (>5s) quality lens (eval) + reel-selection knob ? MEASURED, default-OFF
*(2026-06-18)*
53: rallies longer than `s` *before* matching, consistently across the tIoU and R2 tolerance
paths (a
62:**Measured (n=15 golden, milestone cap6+R4, LOCAL vs golden; reproduce: `python -m
backend.eval.rally_seg_eval --stratify`):**
64:| stratum | court | n | tolF1 | tolP | tolR | tIoU F1@0.5 | seg_ratio |
104:### R1 ? MILESTONE LAUNCHED (cap=6 + R4 flipped default-ON) ? re-verified at n=13
*(2026-06-18)*
105:The R1 milestone is **shipped** (PR #204): the W1 cap=6 + R4 inactivity-end defaults are now
ON.
107:the **[R1 Milestone Launch Report](R1_MILESTONE_LAUNCH_REPORT.md)** (per the Launch-Report
convention).
109:**Re-verified on the grown n=13 corpus** (the 2 new Boxhill clips GX010137/GX010141) ? the
milestone
110:holds and the headline is *stronger* than at n=11:
112:| transition | n=11 (filed) | **n=13 (launched)** |
114:| baseline ? milestone tolF1 | ?+0.232, 10/0, p=0.002 | **?+0.222, 12/0/1, p=0.0005** |
115:| cap12 ? cap6 tolF1 | ?+0.108, 10/0, p=0.002 | **?+0.110, 12/0/1, p=0.0005** |
116:| cap6 ? +R4 tolF1 | ?+0.047, 8/0, p=0.008 | ?+0.040, **9/1/3**, p=0.0215 |
123:W1 cap is the dominant lever / R4 is the marginal +0.040 increment (1 loss = GX010137, a
tolerance
126:``--update-baseline``. W2 / hard-cap / R5 stay default-OFF.
128:### R5 ? blob-fallback for the continuously-active blob (default-OFF, stratum tool)
*(2026-06-17)*
129:The one clip the cap6+R4 milestone still fails is the **continuously-active blob**

```

```

(mahadevpura_1:
130:~174 s of play with no inactivity gap to split on). The W1 gap-split finds no gap; R4 finds
no slow
131:tail; the result is one ~66 s mega-window that matches nothing (**tolF1 0.000**, the
degenerate
132:0-match outcome [ `REAL_FOOTAGE_VALIDATION_REPORT.md` ](REAL_FOOTAGE_VALIDATION_REPORT.md) $5
R5 names).
134:**R5** ( `window_blob_fallback`, default-OFF) is the last resort: **after** the gap-split and
the R4
141:**Why a factor gate, and why not just `window_hard_cap`:** `hard_cap` chops **before** R4
and fires on
142:**every* gap-less over-cap window ? it over-segments normal clips (mahadevpura_2 40?68). R5
runs **after**
143:R4, and the magnitude gate (**default 4x**) leaves a normal long rally (~1?2x the cap)
alone, force-cutting
146:**MEASURED (n=11, on the cap6+R4 milestone, R2 `tolF1`):**
148:| clip | milestone | + R5 | windows (milestone ? +R5) |
156:At factor=4, R5 touches **only** the 2 continuously-active clips + a noise-level BXH2 ?0.004
? effectively
158:mean-swinging 0-match pathology), not a corpus-mean lift (agg `tolF1` 0.323?0.354, sign
**3/3/5, n.s.** at
160:it on selectively for continuously-active footage. The R1 net over-seg guard confirms it:
milestone?+R5 net
163:### R1 ? milestone evidence (cap=6 + R4) + the net over-seg promotion guard *(2026-06-17)*
164:The R1 milestone is the smallest safe local-windowing flip: **W1 bounded windowing cap=6 +
the R4
165:rally-END inactivity trim, no W2** (the per-clip-best config, settled by R3/R4). This entry
(a) re-runs
166:the full ablation ladder baseline?milestone under the **R2** `tolF1` on **n=11** with honest
paired
169:**The full ladder (n=11, R2 `tolF1`, asymmetric ?_start=2/?_end=1.5):**
171:| config | tolF1 (mean [95% CI]) | tIoU-F1 | mean \|?end\| |
175:| + cap=6 (R3) | 0.276 [0.174, 0.380] | 0.407 | 5.7 s |
176:| **+ R4 inact_end=1.5 (milestone)** | **0.323 [0.210, 0.433]** | **0.440** | **3.8 s** |
179:- **baseline ? milestone:** `tolF1` **+0.232** [+0.153, +0.304], **sign 10/0/1, p=0.002**
(the 1 tie is the

```

14. assistant (2026-06-22T01:56:37.096Z)

I now have a very clear, git-grounded picture. The detailed NEXT_STEPS section is stale ? the windowing R-series shipped and the milestone (cap=6 + R4) is the new default-ON state. Let me confirm the cheapest-open-win candidates and read the QUALITY_ITERATIONS frontier (newest entry P7) plus the prominent-court gate to nail down what's genuinely still open and default-OFF.

15. user (2026-06-22T01:56:40.231Z)

=== QUALITY_ITERATIONS top (newest = current frontier, lines 41-75) ===

Current track (live) ? newest first

P7 ? Long-rally (>5s) quality lens (eval) + reel-selection knob ? MEASURED, default-OFF *(2026-06-18)*

Hypothesis (owner). Local OVER-FIRES, and its false positives are mostly SHORT (motion blips, background-court fragments on multicourt). Filtering the eval to **long rallies (>5s)** ? the eye-catching highlights that carry the product ? should strip most short FPs and reveal local's real long-rally quality, answering the North-star "do we beat Gemini on the single-match long-rally happy path?". **Duration** (end?start), NOT shot-count, is the filter: golden `shots\_count` is blank and the no-AI path reports it "Unknown", whereas duration comes from the start/end labels we already trust.

What shipped (both default-OFF, zero behaviour change at the defaults).

- **Eval lens** ? `rally\_seg\_eval.py --min-duration <s>` filters BOTH predictions AND golden GTs

```

to
  rallies longer than `s` *before* matching, consistently across the tIoU and R2 tolerance paths
(a
  FILTER, not a new metric). Plus `--stratify` / `--long-tau` (default 5s): a `{all vs long} ×
{single
  vs multicourt}` aggregate table. Court type = the committed `MULTICOURT_CLIPS` set (the
manifest is
  gitignored, so the set is the auditable source), overridable by a manifest `court_type` field;
kept
  in sync with [`GOLDEN_VIDEOS.md`](data_pipeline/GOLDEN_VIDEOS.md).
- **Product knob** ? `stitching.min_rally_duration` (s, 0 = OFF) drops short rally pairs before
the
  reel compile in `tools/generate_highlights.py` ? a cleaner reel-selection filter than
  `min_shot_count` (which leans on the "Unknown" shot_count).

```

****Measured (n=15 golden, milestone cap6+R4, LOCAL vs golden; reproduce: `python -m backend.eval.rally\_seg\_eval --stratify`)****

	stratum	court	n	tolF1	tolP	tolR	tIoU F1@0.5	seg_ratio
	all	single	11	0.375	0.326	0.464	0.482	1.434
	all	multicourt	4	0.294	0.238	0.393	0.430	1.601
	all	all	15	0.353	0.302	0.445	0.468	1.478
long>5s	single	11	0.264	0.278	0.267	0.416	**1.037**	
long>5s	multicourt	4	0.240	0.197	0.310	0.430	**1.597**	
long>5s	all	15	0.257	0.256	0.279	0.420	1.186	

Pooled counts: single ****608 preds / 416 GTs**** (only 36% of preds are >5s vs 50% of GTs ? long pred:GT ? ****218:210 = 1.04****); multicourt ****288 preds / 179 GTs**** (55% of preds >5s ? long pred:GT = ****158:100 = 1.58****).

```

=== prominent-court gate: default + measured? grep design doc head ===
45:background-court fragments on multicourt). Filtering the eval to **long rallies (>5s)** ? the
55: vs multicourt}` aggregate table. Court type = the committed `MULTICOURT_CLIPS` set (the
manifest is
67:| all | multicourt | 4 | 0.294 | 0.238 | 0.393 | 0.430 | 1.601 |
70:| **long>5s** | multicourt | 4 | 0.240 | 0.197 | 0.310 | 0.430 | **1.597** |
74:**218:210 = 1.04**); multicourt **288 preds / 179 GTs** (55% of preds >5s ? long pred:GT =
**158:100 =
81:2. ?? **Multicourt: thesis FAILS.** Even among >5s rallies local over-fires **1.58x** ?
adjacent-court
83: multicourt; that needs court-localization** (Gate-B / court mask ? gap-closure Lever C).
So "FPs are
100:truths once instrumented: single-court count ?, multicourt ? (needs a court signal), strict-
precision ?
322:**Corpus: golden grew 6 ? 8** this session (`mahadevpura_2` 40 rallies, `GX010128` 52
rallies ? both real multicourt,
590:> retired from the box (their labels are gone); the present golden set is **9 multicourt
clips**. On that

```

16. user (2026-06-22T01:56:43.233Z)

```

=== person fusion P3 measured result on golden? (the cheapest-vs-deferred question) ===
12:graded **offline against golden labels** (per-video + LOVO-honest); promotion to default
happens **only
14:`git revert`.** Cues are designed in
[MULTI_SIGNAL_FUSION_PLAN.md](MULTI_SIGNAL_FUSION_PLAN.md); scoring
18:**Golden corpus = the 15 labelled matches** in `output/human_lovo_manifest.json` (the
original 6 ?
30:| Gen-0 owned head (LOVO, n=6) | 0.744 | 0.561 |
37:is the precision bug the fusion gates target.
120:milestone 0.474. Independently reproduced by a 2nd harness (#214 `lovo_resweep.py`, matches
to 3 dp;

```



```

121:LOVO picks cap=6 **13/13** on tIoU) and survived a 5-lens adversarial workflow (every L00
fold clears
241:A scheduled nightly job re-scores the **golden corpus** (`output/human_lovo_manifest.json`,
n=11) under HEAD
323:frame-aligned, added to `human_lovo_manifest.json`). **n=8 milestone numbers (LOVO,
tIoU-F1):** baseline 0.232 ? W1
337:**MEASURED ? golden n=6 (W1 cap=12 ? +hard-cap):** ?F1 **+0.006** [?0.019, +0.033], **CI
spans 0,
363:(`FusionSegmenter._select_for_handoff`, served; `rally_seg_eval.windows_for`, eval) now fall
back to the
373:### Tier 1 (W2) ? net-crossings rally-state gate on W1: ?F1 +0.019, 6/0 *(2026-06-16)*
379:gate". The gate itself is **existing code** (`FusionSegmenter.min_crossings`); served path
gets W1+W2 via config
380:(`indexing.max_window_duration: 12` on `fusion_hybrid` + `indexing.fusion.min_crossings: 1`)
? no new detector code.
382:**MEASURED on golden (n=6, LOVO; W1 cap=12, then + net-crossings ?1):**
395:Default-OFF; promote via config. The richer multi-feature scorer (person/audio + direction-
reversal/peak) is the W2
396:follow-on (needs GPU person/audio feature re-extraction on the golden videos ? now feasible
with the GPU grant).
398:### Tier 1 (W1) ? bounded windowing fixes the over-merge: ?F1 +0.139, 6/0 *(2026-06-16)*
408:**MEASURED on golden (n=6, LOVO, `rally_seg_eval --vs`, recommended cap=12s /
min_split_gap=0.5):**
454:### Served-side Gate-A measurement on the LIVE FusionSegmenter ? the independent run behind
the #151 revert *(2026-06-14)*

```

=== Is there a 'next' / open-frontier marker in MULTI_SIGNAL_FUSION_PLAN status header? ===

Multi-signal rally fusion ? add a human (person) cue alongside the shuttle

```

**Status:** **P1 DONE** (2026-06-13) ? the person cue is extracted to
`backend/pipeline/detectors/person_detector.py` (`PersonDetector`), consumed by `yolo_hybrid`
with no
behavior change. **P2?P4 remain owner-gated.** Phases below are sequenced but each is a separate
PR.
**Owner decisions captured** in [S9](#9-open-decisions-for-the-owner).

```

```

> **Companion docs:** the 2-layer mental model is
[HOW_RALLY_DETECTION_WORKS.md](HOW_RALLY_DETECTION_WORKS.md);
> the modular-fusion architecture this builds on is **ADR-007**
([archives/decisions/DECISIONS.md](archives/decisions/DECISIONS.md));
> the permissive-dependency rule is **ADR-002 / ADR-005**; the audio cue (the #1 next cue,
already
> scaffolded) is [archives/research/AUDIO_RALLY_DETECTION_PLAN.md](archives/research/AUDIO_RALLY
_DETECTION_PLAN.md).
> The vendor labels that gate the learned path come from
[GOLDEN_SET_VENDORING.md](archives/roora-vendor/GOLDEN_SET_VENDORING.md)
> and the labeling spec in [ROORA_LABELING_GUIDELINES.md](archives/roora-
vendor/ROORA_LABELING_GUIDELINES.md).
> **How each cue here is added/tested without production regression** ? A/B, shadow eval, and
the
> promote-only-on-lift discipline ? is [EXPERIMENT_HARNESS.md](EXPERIMENT_HARNESS.md) (every
P1?P4 cue
> below lands flag-gated, default OFF, promoted only on measured LOVO lift).

```

1. Context ? why this, why now

Rally detection today runs **one cue at a time**. Each segmenter independently turns a single signal into rally windows:

```

| Segmenter | Signal | File |
|---|---|---|

```

```
| `wasb_hybrid` / `tracknet_hybrid` | shuttle velocity (frozen WASB/TrackNet trajectory) |  
`backend/pipeline/detectors/tracknet_runner.py` |  
| `yolo_hybrid` (the Pydantic `default_segmenter` fallback; `config.json` overrides it to  
`gemini`) | person velocity (torchvision + ByteTrack) |  
`backend/pipeline/segmenters/yolo_hybrid.py` |  
| `motion` | raw pixel motion | `backend/pipeline/segmenters/motion.py` |  
| `gemini` | a cloud VLM watching the whole clip | `backend/providers/gemini.py` |
```

17. assistant (2026-06-22T01:56:48.974Z)

I now have everything needed. Let me do one final check on the most actionable open item ? the prominent-court gate (#243, default-OFF, the #1 measured remaining gap = multicourt FPs) ? to confirm it's measured-but-not-promoted, which determines whether it's the cheapest win.

18. user (2026-06-22T01:56:52.965Z)

```
=== prominent-court gate config knob + default (the multicourt FP fix #243) ===  
origin/master:backend/eval/rally_seg_eval.py:162:      # Optional prominent-court gate inputs  
(multicourt G2; default absent ? the gate is a no-op):  
origin/master:backend/eval/rally_seg_eval.py:190:      # court_polygon, drop candidate rallies  
whose shuttle is MAJORITY outside the prominent court ? a  
origin/master:backend/pipeline/detectors/court_gate.py:5:shuttle signal `(x, y, visible)` ?  
`(x, y, visible AND inside-the-prominent-court)` : a visible  
origin/master:backend/pipeline/detectors/court_gate.py:6:shuttle point whose normalised position  
is outside the prominent-court polygon is marked NOT  
origin/master:backend/pipeline/detectors/court_gate.py:18:admitted, so a high clear over the  
prominent court is not clipped.  
origin/master:backend/pipeline/detectors/court_gate.py:35:      over the prominent court, admit  
it. """  
origin/master:backend/pipeline/detectors/court_gate.py:44:      """Fraction of a candidate rally's  
VISIBLE shuttle points that lie inside the prominent court.  
origin/master:backend/pipeline/detectors/court_gate.py:69:      occasionally drifts into the  
prominent court keeps a few in-court points, enough to hold a (sparse)  
origin/master:backend/pipeline/detectors/court_gate.py:84:      """Return ``points`` with  
visibility CLEARED for any visible shuttle outside the prominent-court  
  
=== Was prominent-court MEASURED on golden, or just built? ===  
--- look for a measured entry ---
```

19. assistant (2026-06-22T01:56:58.192Z)

The prominent-court gate (#243) is **built** and wired into eval but NOT yet measured on golden** (no measured entry in QUALITY_ITERATIONS, requires a `court\_polygon` input that's absent by default = no-op). This is the gate targeting the #1 confirmed remaining gap (multicourt background-court FPs, the P7 thesis that "FAILS" at 1.58× over-fire).

I have a complete, git-grounded picture. Let me compile the structured output.

20. user (2026-06-22T01:57:40.846Z)

Structured output provided successfully

21. assistant (2026-06-22T01:57:50.881Z)

I've completed the git-grounded scan of the rally-detection quality track. Summary of what I found and verified against engine origin/master (not just docs):

****The cheapest OPEN quality win**** is now the ****prominent-court gate (#243)**** ? it's already built (`backend/pipeline/detectors/court\_gate.py`) and wired into `rally\_seg\_eval.py` as a default-OFF no-op, but has no measured golden entry yet. It targets the #1 confirmed remaining gap: multicourt background-court false positives (the P7 thesis "FPs are short" explicitly FAILS on multicourt at 1.58× over-fire). It just needs court polygons + an A/B measurement ? promote on lift.

****Key stale-doc corrections**** (the anti-spiral catches):

- The Gate-A wiring / FusionSegmenter / multi-signal fusion P1 the doc lists as "next" all ****shipped****. The Gate-A served-default flip (#146) was tried, ****regressed****, and was ****reverted** (#151) **** ?** that loop is closed; Gate A is correctly opt-in.
- The R-series ****all shipped****: R2 metric (#199), R4 end-rule (#200), and critically the ****R1** milestone flipped windowing defaults ON ****** (#204, cap=6 + R4, re-verified at n=13, p=0.0005). That **** ?** not Gate A **** ?** is the dominant quality lever and it's already in production defaults.
- The "wire Gate A = cheapest win" framing in NEXT_STEPS is stale; the cheapest ***open*** win moved to the prominent-court gate.

Everything is captured in the StructuredOutput call with priorities, effort, and owner-gating. The two genuinely-open, near-term levers (prominent-court measurement + person-cue LOVO) are both blocked on owner-side GPU golden features; corpus growth (P0) remains the highest-leverage owner action.